

Bonus 3 - RESEM' (PREZ.)

① a) $y' = (12x^3 - 3) \cdot \ln x + (3x^4 - 3x + 5) \cdot \frac{1}{x}$

b) $y' = \frac{3x^2}{2\sqrt{x^3+1}} - 1$

② a) $z'_x = 5x^4 y^2 - 4y^3 + e^x$

$$z'_y = 2x^5 y - 12xy^2$$

b) $z'_x = y \cdot \cos(xy^2) \cdot y^2 = y^3 \cdot \cos(xy^2)$

$$z'_y = \sin(xy^2) + y \cdot \cos(xy^2) \cdot 2xy$$

$$= \sin(xy^2) + 2xy^2 \cdot \cos(xy^2)$$